

Mutual Fund Analytics Capstone Project Report

Executive Summary

The Mutual Fund Analytics project was developed to analyze mutual fund performance, investor behavior, and risk metrics using historical financial datasets. The primary objective was to build an end-to-end analytics pipeline covering ETL, SQL-based storage, exploratory data analysis, performance evaluation, advanced risk analytics, and dashboard visualization.

The project processed multiple datasets including NAV history, AUM data, SIP inflows, category inflows, investor transactions, and benchmark indices. By integrating these datasets, meaningful insights were generated regarding fund growth, investor preferences, and risk-return profiles.

The final output includes a Power BI dashboard, Python notebooks, SQL database design, advanced analytics models, and a recommendation system.

Problem Statement

Mutual fund investors often face challenges in selecting funds due to the complexity of analyzing returns, volatility, and risk metrics. Traditional fund analysis is scattered across multiple reports and platforms.

This project aims to centralize and simplify mutual fund analysis by building a complete analytics solution capable of:

- Tracking fund performance
- Measuring risk-adjusted returns
- Understanding investor behavior
- Generating actionable insights

- Supporting fund recommendations
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Project Objectives

The major objectives of the project are:

1. Build an ETL pipeline for mutual fund datasets.
 2. Clean and structure the data for analysis.
 3. Design an SQL database for efficient storage.
 4. Perform exploratory data analysis.
 5. Compute advanced performance metrics.
 6. Measure portfolio risk using statistical models.
 7. Develop a Power BI dashboard.
 8. Create a simple mutual fund recommendation system.
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Data Sources

The project used the following datasets:

1. NAV History

Contains daily NAV values for 40 mutual fund schemes from 2022 to 2026.

2. AUM Data

Tracks Assets Under Management across fund houses.

3. SIP Data

Monthly SIP inflows and account growth.

4. Category Inflows

Monthly net inflows into equity, debt, and hybrid categories.

5. Investor Transactions

Contains investor transaction records with demographic information.

6. Portfolio Holdings

Contains stock-wise sector allocations of mutual funds.

7. Benchmark Indices

Includes NIFTY50 and NIFTY100 daily closing values.

ETL Design

The ETL pipeline was divided into three major phases:

Extract

Raw CSV files were collected and loaded into Python.

Transform

Data cleaning included:

- Removing null values
- Fixing date formats
- Standardizing categories
- Handling duplicates

Load

Cleaned data was loaded into SQLite database tables.

Database tables created:

- dim_fund
- fact_nav

- fact_aum
 - fact_sip
 - fact_category_inflows
 - fact_folio
 - fact_transactions
 - fact_holdings
 - fact_benchmark
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Exploratory Data Analysis

NAV Trend Analysis

NAV trends showed consistent growth across most funds, with strong upward movement during bullish periods and temporary corrections during market volatility.

AUM Growth

SBI Mutual Fund emerged as the leading fund house by AUM.

SIP Inflow Analysis

SIP inflows showed continuous growth, reaching all-time highs in late 2025.

Category Inflow Heatmap

Equity funds attracted the highest inflows compared to debt and hybrid funds.

Investor Demographics

The 26–35 age group represented the largest investor segment.

Geographic Distribution

T30 cities contributed the majority of investments.

Folio Growth

Total folio count nearly doubled over the analysis period.

Performance Analytics

Several performance metrics were computed:

CAGR

CAGR was calculated for all schemes to measure annualized growth.

Sharpe Ratio

Sharpe ratio identified funds delivering better returns per unit of risk.

Sortino Ratio

Sortino ratio focused on downside volatility.

Alpha

Alpha measured excess return over benchmark.

Beta

Beta measured fund sensitivity to benchmark movements.

Maximum Drawdown

Maximum drawdown measured worst historical decline.

Fund Scorecard

A weighted ranking model was built using:

- 30% Return Rank
- 25% Sharpe Ratio
- 20% Alpha

- 15% Expense Ratio
 - 10% Drawdown
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Advanced Risk Analytics

Value at Risk (VaR)

Historical VaR at 95% confidence measured downside risk.

Conditional VaR (CVaR)

CVaR measured expected loss beyond the VaR threshold.

Rolling Sharpe

Rolling 90-day Sharpe analysis showed changing fund stability.

Investor Cohort Analysis

Cohort analysis grouped investors based on first investment year.

SIP Continuity Analysis

Investors with irregular SIP patterns were identified.

Sector Concentration (HHI)

HHI measured sector concentration in fund portfolios.

Dashboard Overview

The Power BI dashboard contains:

1. NAV trend visualizations
2. Fund performance scorecards
3. SIP inflow trends

4. AUM comparison
5. Investor demographics
6. Risk analytics charts
7. Benchmark comparison

Dashboard provides a consolidated view for fund analysis.

(Add dashboard screenshots here)

Key Findings

1. SBI Mutual Fund dominates AUM market share.
 2. Equity funds attracted maximum investor interest.
 3. SIP inflows reached record highs in 2025.
 4. Younger investors are driving mutual fund growth.
 5. T30 cities contribute most investments.
 6. Top-performing funds show strong Sharpe ratios.
 7. High-alpha funds outperform benchmark consistently.
 8. VaR analysis identifies high-risk funds clearly.
 9. SIP discontinuity can indicate investor churn.
 10. Sector concentration varies significantly across funds.
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Limitations

1. Dataset is historical and static.
 2. No live market integration.
 3. Recommendation engine is basic.
 4. Limited macroeconomic factors included.
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Recommendations

1. Integrate live NAV APIs.
2. Add real-time benchmark updates.

3. Improve recommender using ML models.
 4. Add investor personalization features.
 5. Expand dashboard deployment to cloud.
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Conclusion

The Mutual Fund Analytics project successfully demonstrates how financial datasets can be transformed into actionable business insights using Python, SQL, and Power BI.

The project provides a strong foundation for advanced financial analytics, risk modeling, and investor-focused recommendation systems. It bridges the gap between raw market data and decision-making by offering a structured analytical framework.

This capstone strengthened skills in ETL, data engineering, analytics, visualization, and financial modeling.